

CONVERSATIONAL CLUSTERING

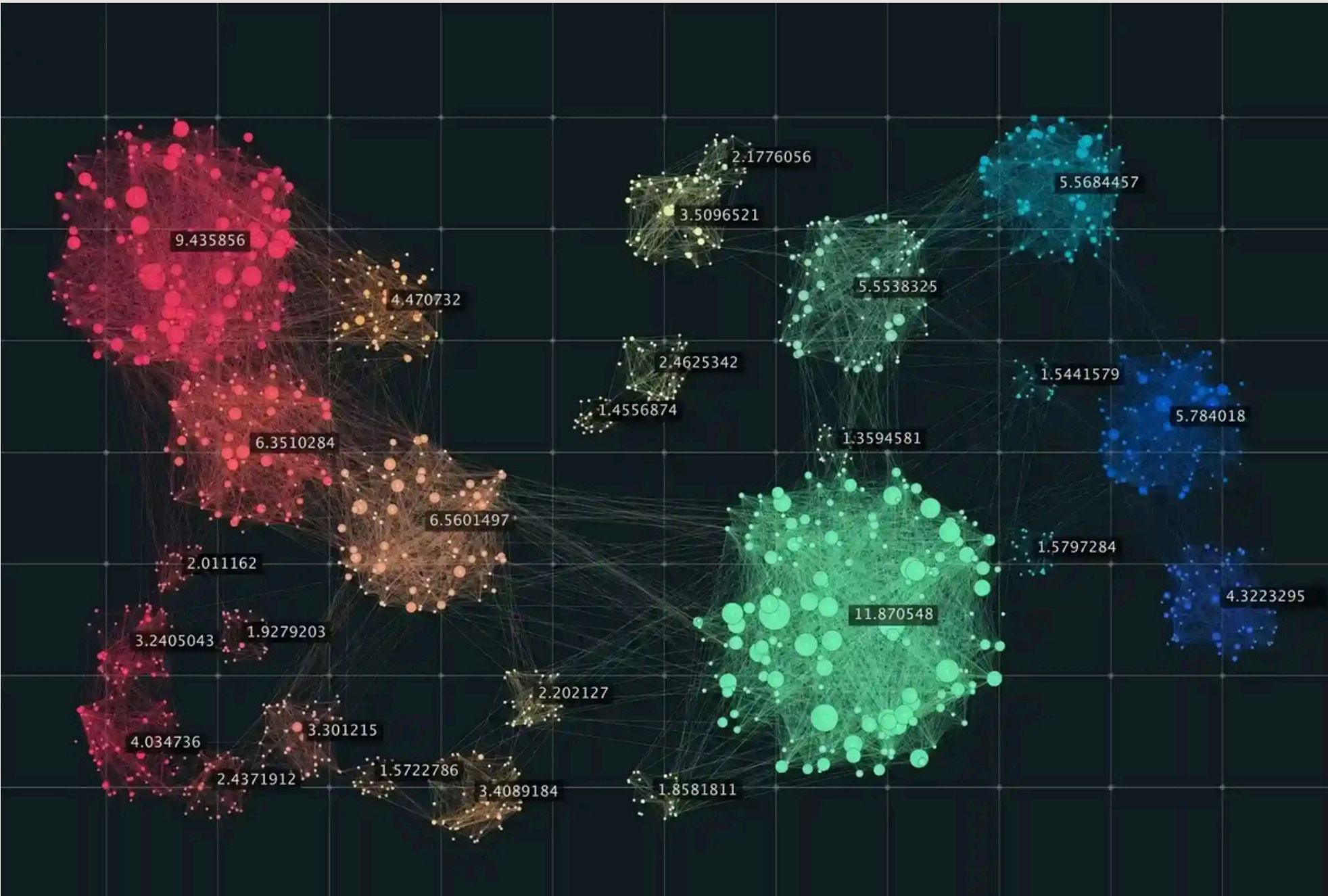
PROBLEM STATEMENT

TECHNICAL KNOWLEDGE

Users often lack technical knowledge about clustering techniques

FINETUNING

Translating high-level intents into technical choices places the cognitive burden in the wrong place



CLAIM

Conversational clustering allows users to easily express their preferences, and at the same time achieve comparable or better results than using classical clustering algorithms (manually fine-tuned).

QUESTION 1

Does a conversational clustering system allow non-expert users to produce cluster assignments of comparable quality to those produced by expert users of classical clustering tools (e.g. K-means with manual hyperparameter tuning), while requiring less domain-specific technical knowledge?

QUESTION 2

Given that the conversational interface works, what properties of the conversation affect how quickly and faithfully the user's intent is captured?
Specifically: does question type (open-ended / binary pairwise / example triplet), conversation length (number of turns), and total tokens exchanged predict preference-alignment score?

OTHER QUESTIONS

- Will the system be able to process large datasets (e.g. hundreds of thousands or more datapoints)?
- Can/does the system work on non-text data (tabular, image, time-series)? The current design assumes text input throughout
- How does the system behave when the user has no strong clustering preference and cannot answer the LLM's questions meaningfully?
- How robust is the system to vague, ambiguous, or deliberately unhelpful user answers?
- Can the system update the clustering as new data arrives, without restarting the conversation?
- Can the system explain why a specific datapoint was assigned to a specific cluster in terms the user understands?

RELATED WORK: CONVERSATION

RELEVANT PAPERS

- Eliciting Human Preferences with Language Models
- TO-GATE: Clarifying Questions and Summarizing Responses with Trajectory Optimization for Eliciting Human Preference
- **Asking Clarifying Questions for Preference Elicitation With Large Language Models**

RELATED WORK: EMBEDDINGS

RELEVANT PAPERS

- Improving Text Embeddings with Large Language Models

RELATED WORK: CLUSTERING

RELEVANT PAPERS

- Large Language Models Enable Few-Shot Clustering
- Dial-In LLM: Human-Aligned LLM-in-the-loop Intent Clustering for Customer Service Dialogues
- Optimized Algorithms for Text Clustering with LLM-Generated Constraints
- Summaries as Centroids for Interpretable and Scalable Text Clustering

ARCHITECTURE

CONVERSATION

The user explains its clustering preferences to the LLM. If needed, clarifying questions are asked by the model, to better frame the user's intentions

EMBEDDINGS GENERATION

A pre-trained encoder (such as e5-mistral-7b-instruct) is used to encode all the documents in the dataset. The user's preferences are also taken into account.

CLUSTERING

A clustering algorithm is run on the embedded documents. Strategies such as summary-as-centroids or keyphrase generation can be used to improve the results.

DATASETS

STACKOVERFLOW

Around 20k short texts (question titles).
Clustering can be based on programming language, architectural vs debugging questions, etc.

GOEMOTIONS

Around 58k Reddit comments.
Clustering can be based on the emotion of the comments.

EVALUATION

CLUSTERING: INTERNAL METRICS

Silhouette coefficient (or similar)

CLUSTERING: EXTERNAL METRICS

Comparison with a ground-truth may be misleading

CONVERSATION: UTILITY

The utility of a conversation is evaluated based on the number of tokens needed, turns and questions needed